

OUTPUT :

1. Adam is working in an IT company. He has been given a task to reduce the load of a system by killing some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation?

- **Kill Processes by name**

```
vaishnavi@DESKTOP-B9LM4V1: ~  
root@DESKTOP-B9LM4V1:~# su - vaishnavi  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef  
UID          PID    PPID  C STIME TTY          TIME CMD  
root           1         0  0 19:51 ?           00:00:00 /init  
root        288         1  0 20:21 tty1          00:00:00 /init  
root        289       288  0 20:21 tty1          00:00:00 -bash  
root        304       289  0 20:21 tty1          00:00:00 su - vaishnavi  
vaishna+     305       304  1 20:21 tty1          00:00:00 -bash  
vaishna+     320       305  0 20:21 tty1          00:00:00 ps -ef  
vaishnavi@DESKTOP-B9LM4V1:~$ pkill vaishnavi  
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef|vaishnavi  
vaishnavi: command not found  
vaishnavi@DESKTOP-B9LM4V1:~$
```

- **Kill a process based on the process name**

```
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef  
UID          PID    PPID  C STIME TTY          TIME CMD  
root           1         0  0 19:51 ?           00:00:00 /init  
root        288         1  0 20:21 tty1          00:00:00 /init  
root        289       288  0 20:21 tty1          00:00:00 -bash  
root        304       289  0 20:21 tty1          00:00:00 su - vaishnavi  
vaishna+     305       304  0 20:21 tty1          00:00:00 -bash  
vaishna+     331       305  0 20:23 tty1          00:00:00 ps -ef  
vaishnavi@DESKTOP-B9LM4V1:~$ sleep 300 & [1] 983  
[1] 332  
[1]: command not found  
vaishnavi@DESKTOP-B9LM4V1:~$ kill 332  
[1]+  Terminated                  sleep 300  
vaishnavi@DESKTOP-B9LM4V1:~$
```

- Kill a single process at a time with the given process ID

vaishnavi@DESKTOP-B9LM4V1: ~

```
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 19:51 ?           00:00:00 /init
root         288          1  0 20:21 tty1       00:00:00 /init
root         289        288  0 20:21 tty1       00:00:00 -bash
root         304        289  0 20:21 tty1       00:00:00 su - vaishnavi
vaishna+     305        304  0 20:21 tty1       00:00:00 -bash
vaishna+     331        305  0 20:23 tty1       00:00:00 ps -ef
vaishnavi@DESKTOP-B9LM4V1:~$ sleep 300 & [1] 983
[1] 332
[1]: command not found
vaishnavi@DESKTOP-B9LM4V1:~$ kill 332
[1]+  Terminated                  sleep 300
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 19:51 ?           00:00:00 /init
root         288          1  0 20:21 tty1       00:00:00 /init
root         289        288  0 20:21 tty1       00:00:00 -bash
root         304        289  0 20:21 tty1       00:00:00 su - vaishnavi
vaishna+     305        304  0 20:21 tty1       00:00:00 -bash
vaishna+     341        305  0 20:26 tty1       00:00:00 ps -ef
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef |grep sleep
vaishna+    343        305  0 20:26 tty1       00:00:00 grep --color=auto sleep
vaishnavi@DESKTOP-B9LM4V1:~$ killall sleep
sleep: no process found
vaishnavi@DESKTOP-B9LM4V1:~$ Killall sleep
Command 'Killall' not found, did you mean:
  command 'killall' from deb psmisc (23.4-2build3)
Try: sudo apt install <deb name>
vaishnavi@DESKTOP-B9LM4V1:~$
```

2 . Write a program for process creation using C .

- **Orphan Process**

```
compilation terminated.
vaishnavi@DESKTOP-B9LM4V1:~$ ls
Zombie.save  orphan.c.save  orphan.c.save.2  zoombie
orphan.c      orphan.c.save.1  zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ nano orphan.c
vaishnavi@DESKTOP-B9LM4V1:~$ gcc orphan.c -o orphan
vaishnavi@DESKTOP-B9LM4V1:~$ gcc orphan.c -o orphan./orphan
/usr/bin/ld: cannot open output file orphan./orphan: No such file or directory
collect2: error: ld returned 1 exit status
vaishnavi@DESKTOP-B9LM4V1:~$ gcc orphan.c -o orphan
vaishnavi@DESKTOP-B9LM4V1:~$ ./orphan
Parent Process exiting
vaishnavi@DESKTOP-B9LM4V1:~$ cChild Process
PID: 1645
PPID: 1

vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef|grep orphan
vaishnavi+ 1647 1519 0 01:19 tty1      00:00:00 grep --color=auto orphan
vaishnavi@DESKTOP-B9LM4V1:~$
```

- **Zombie Process**

```
Zombie.save: Command not found
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ ls
Zombie.save  orphan.c.save  orphan.c.save.1  orphan.c.save.2  zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ gcc Zombie.c -o zoombie
cc1: fatal error: Zombie.c: No such file or directory
compilation terminated.
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ ls
Zombie.save  orphan.c.save  orphan.c.save.1  orphan.c.save.2  zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ gcc zombie.c -o zoombie
vaishnavi@DESKTOP-B9LM4V1:~$ ./zombie
-bash: ./zombie: No such file or directory
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ ls
Zombie.save  orphan.c.save.1  zombie.c
orphan.c.save  orphan.c.save.2  zoombie
vaishnavi@DESKTOP-B9LM4V1:~$ gcc zombie.c -o zoombie
vaishnavi@DESKTOP-B9LM4V1:~$ ./zombie
-bash: ./zombie: No such file or directory
vaishnavi@DESKTOP-B9LM4V1:~$ ./zoombie
Child process exiting
Parent process running
vaishnavi@DESKTOP-B9LM4V1:~$ ps -el|grep z
vaishnavi@DESKTOP-B9LM4V1:~$ ps -el|grep Z
F S  UID  PID  PPID  C PRI  NI ADDR SZ WCHAN  TTY      TIME CMD
vaishnavi@DESKTOP-B9LM4V1:~$
```

3. Create the process using fork () system call.

- **Child Process creation**
- **Parent Process creation**

```
vaishnavi@DESKTOP-B9LM4V1:~$ nano fork.c
vaishnavi@DESKTOP-B9LM4V1:~$ ./fork
-bash: ./fork: No such file or directory
vaishnavi@DESKTOP-B9LM4V1:~$ nano fork.c
vaishnavi@DESKTOP-B9LM4V1:~$ gcc fork.c -o fork
vaishnavi@DESKTOP-B9LM4V1:~$ ./fork
Parent Process
Child Process
PID: 1666
PID: 1667
Child PID: 1667
PPID: 1666
vaishnavi@DESKTOP-B9LM4V1:~$
```

PPID and PID

```
vaishnavi@DESKTOP-B9LM4V1: ~
root@DESKTOP-B9LM4V1:~# su - vaishnavi
vaishnavi@DESKTOP-B9LM4V1:~$ sleep 3008 [1] 2754
sleep: invalid time interval '[1]'
Try 'sleep --help' for more information.
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef|grep sleep
vaishna+ 1674 1551 0 01:49 tty2 00:00:00 grep --color=auto sleep
vaishnavi@DESKTOP-B9LM4V1:~$ kill 1674
-bash: kill: (1674) - No such process
vaishnavi@DESKTOP-B9LM4V1:~$ killall sleep
sleep: no process found
vaishnavi@DESKTOP-B9LM4V1:~$
```